

Notes: Bernstein, Korteweg & Laws (JF, 2017)

Attracting Early-Stage Investors: Evidence from a Randomized Field Experiment

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1 Big picture

- **Question.** Which characteristics of early-stage firms causally attract investor attention: founding team, traction, or existing investors?
- **Setting.** AngelList featured emails sent to accredited investors about real start-ups seeking funding.
- **Experiment.** Investors are randomly shown or not shown information categories that would normally pass AngelList disclosure thresholds.
- **Main result.** Investors respond strongly to team information; showing team information increases click-through by about 2.2–2.3 percentage points.
- **Nonresult.** Current-investor and traction disclosures are not significant in randomized specifications.
- **Economic takeaway.** Human capital assets are causally important in very early-stage financing.

2 Incremental contribution of the paper

- **First causal field evidence on early-stage investor screening.** Prior evidence was mainly surveys, interviews, or observational correlations.
- **Direct test of competing theories.** The design contrasts human-capital, traction/nonhuman-asset, and information-cascade views.
- **Within-start-up randomization.** The experiment solves the omitted-variable problem that team, traction, and lead investors usually co-move with latent quality.
- **Real market behavior.** The study uses actual AngelList emails, real start-ups, and active accredited investors.
- **Mechanism evidence.** Specialized and experienced-investor responses suggest that team information is not merely a signal about the idea but an operationally relevant asset.

3 Data and measurement

3.1 AngelList platform and experiment

AngelList is an online platform matching early-stage start-ups with accredited investors. Start-ups post product descriptions, team backgrounds, traction, current investors, fundraising goals, and terms. The field experiment randomizes disclosure of three categories in featured emails: team, current investors, and traction.

Interpretation. The platform is useful because it reveals both the information investors receive and their immediate response.

3.2 Randomized categories and outcomes

The three categories are randomly hidden or shown conditional on passing an AngelList disclosure threshold. Outcomes include opened email, click on the View button, introduction request, and observed investment. The main regression sample is opened emails.

Interpretation. The causal object is the effect of revealing a category, not the effect of having that characteristic.

3.3 Descriptive statistics

The experiment sends 16,981 emails to 4,494 active investors. 2,925 investors open at least one email, and the main regression sample contains 8,189 opened emails. The experiment covers 21 featured start-ups.

Interpretation. The sample is selected but economically meaningful: it consists of real early-stage financing opportunities shown to active investors.

4 Empirical specifications

4.1 Baseline randomized response regression

$$Clicked_{is} = \beta_T TeamShown_{is} + \beta_I InvestorShown_{is} + \beta_R TractionShown_{is} + \gamma Controls_{is} + \mu_s + \varepsilon_{is}.$$

Interpretation. Start-up fixed effects mean identification comes from different investors receiving different versions of the same start-up email.

4.2 Specialization and experience heterogeneity

$$Clicked_{is} = \beta CategoryShown_{is} + \eta(CategoryShown_{is} \times Type_i) + Controls_{is} + \mu_s + \varepsilon_{is}.$$

Interpretation. Interactions test whether specialized, experienced, or successful investors respond differently to team, traction, or current-investor information.

5 Identification strategy

5.1 Causal estimand: disclosure effects conditional on eligibility

The experiment does not estimate the value of having a good team, having customer traction, or having high-quality existing investors in the population of all start-ups. It estimates a more precise causal object: the effect on investor attention of *revealing* an information category in the featured email, conditional on that category having already passed AngelList’s disclosure threshold.

$$\beta_k = E[Y_{is}(1_k) - Y_{is}(0_k) \mid \text{category } k \text{ passed the platform threshold}],$$

where k indexes team, current-investor, or traction information, and Y_{is} is an investor response to start-up s .

Interpretation. This distinction is important. A non-significant traction coefficient does not mean traction is worthless. It means that, within this email format and among firms whose traction was strong enough to be eligible for disclosure, showing the traction line did not causally increase the average opened-email click rate. The paper is therefore a test of *marginal information disclosure*, not a complete production function for start-up value.

5.2 Within-start-up randomization

The core identification comes from random variation in what investors see about the same start-up. For a given featured company, investors receive the same common description, funding context, and baseline

email layout, but the experiment randomizes whether eligible team, current-investor, and traction categories appear.

- Start-up fixed effects absorb the underlying idea, industry, geography, financing target, valuation, product description, and all common quality components of the start-up.
- The coefficients are identified from comparisons across investors who view different information versions of the same start-up email.
- Because the same start-up appears in treatment and control versions, omitted firm quality cannot mechanically explain the randomized category coefficients.

Interpretation. The design solves the central problem in observational entrepreneurial-finance data: strong teams, strong traction, and strong backers usually move together. Randomization breaks this co-movement at the moment of investor screening.

5.3 Opened-email sample and interpretation of the outcome

The baseline analysis uses opened emails, because only investors who open the email can observe the randomized information categories. The main outcome is whether the investor clicks the “View” button in the email.

- The opened-email restriction makes the treatment meaningful: unopened emails cannot transmit team, investor, or traction information.
- Clicks are an attention and screening outcome, not a final investment outcome.
- Clicks matter economically because they can lead to introduction requests, and introductions are the platform’s next step toward investment.
- Observed investments are harder to use as the main outcome because investments are less frequent, underreported, and often occur after unobserved offline interactions.

Interpretation. The cleanest causal outcome is investor attention. The paper is strongest about what attracts early-stage investor screening interest; it is more cautious about long-run investment returns or final entrepreneurial success.

5.4 Regression controls and fixed effects

The baseline specification includes start-up fixed effects and, in the controlled columns, variables such as investor-start-up connections, whether the investor already followed the start-up, and the number of prior featured emails received.

$$Clicked_{is} = \beta_T TeamShown_{is} + \beta_I InvestorShown_{is} + \beta_R TractionShown_{is} + \gamma' X_{is} + \mu_s + \varepsilon_{is}.$$

- μ_s absorbs all common start-up-level information.
- X_{is} controls for investor-start-up network exposure and investor familiarity.
- Investor-clustered standard errors allow repeated observations for the same investor.

Interpretation. The treatment assignment is randomized, so the controls are not needed for identification in principle. They improve precision and show that the team effect is not driven by preexisting connections or repeated exposure.

5.5 Why the team coefficient is interpreted causally

A positive β_T means that investors who see the team line for a given start-up are more likely to click than investors who do not see the team line for the same start-up. Since the disclosure was randomized, the coefficient is not contaminated by the possibility that stronger teams are attached to better ideas.

Interpretation. This is the paper’s main causal contribution. Observationally, team quality may proxy for hidden start-up quality. Experimentally, the same hidden quality is held fixed, and only the visibility of team information changes.

5.6 Why current-investor and traction nonresults are informative

The non-significant randomized coefficients for current investors and traction are informative because they are estimated in the same experimental design as the team coefficient.

- If investors mainly relied on information cascades, current-investor disclosure should have raised clicks.
- If investors mainly relied on early nonhuman assets or customer validation, traction disclosure should have raised clicks.
- The fact that team disclosure moves clicks more strongly than the other categories helps rank the importance of alternative screening theories in this setting.

Interpretation. The result is not that backers and traction never matter. It is that, in this early-stage featured-email environment, the causal attention response is strongest for human-capital information.

5.7 Mechanism and heterogeneity tests

The paper uses heterogeneity by investor specialization, experience, and prior success to interpret why team information matters.

- Specialized investors responding to team information suggests that the team is not merely a crude proxy for the business idea; specialized investors should already understand the relevant product or sector.
- More experienced and more successful investors responding mainly to team information suggests that selecting on teams may be a rational screening strategy rather than an inexperienced-investor behavioral bias.
- Inexperienced investors responding to traction and current-investor information suggests those categories do contain information, even if experienced investors choose to focus elsewhere.

Interpretation. These tests do not prove that team-based investing generates superior returns, but they strengthen the interpretation that team quality is operationally meaningful and not only a decorative signal in the pitch.

5.8 Threats to identification and how the paper addresses them

- **Category-order concern.** The email format always lists information in the same order. This avoids suspicious changes to AngelList’s normal format but means the paper cannot separately estimate presentation-order effects. The key comparison remains randomized show/hide within the existing order.
- **Disclosure-threshold concern.** Categories are randomized only after passing AngelList thresholds. The paper therefore compares disclosure effects among above-threshold information, not low-quality undisclosed information.

- **Signal-to-noise concern.** If the traction or current-investor thresholds were poorly chosen, their nonresults could reflect weak signal quality. The paper argues against a purely mechanical threshold explanation by showing that inexperienced investors do respond to these categories.
- **Spillover concern.** Investors could potentially learn about start-ups outside the email. Prior-follow and connection controls help, and randomization still ensures that such exposure is not mechanically aligned with treatment assignment.
- **External-validity concern.** Featured firms are selected by AngelList and investors are active accredited investors. Table X addresses this by comparing experiment firms to a broader AngelList sample.
- **Outcome limitation.** Clicks are not final investment outcomes. The paper interprets clicks as screening interest, supported by conversion from clicks to introductions, but it avoids claiming direct causal effects on realized firm success.

Interpretation. The paper has very strong internal validity for investor attention within the AngelList featured-email setting. Its main limits are external validity and the distance between initial investor attention and long-run entrepreneurial outcomes.

5.9 External validity and scope

The experiment covers real start-ups and real accredited investors, but the companies are selected for featured emails and the investors are active AngelList users. The broader-sample comparison in Table X shows that featured firms are somewhat stronger on some observable dimensions, such as founder count, employees, advisors, and incubator history, but not radically different on every dimension.

Interpretation. The most defensible external-validity statement is: the results apply best to serious seed-stage firms that are visible enough to be shown to active angel investors on an online platform. The paper is less directly informative about later-stage VC deals, offline founder-investor relationships, or start-ups too weak to be featured.

6 Tables and figures

6.1 Figure 1: Sample featured start-up email to investors

Read: Shows the actual email format and the randomized categories.

Economic message. The figure clarifies how the experiment isolates information salience.

6.2 Table I: Descriptive Statistics of Emails for Randomized Field Experiment

Read: 16,981 emails are sent; opened email rate is 48.22%; click rate among opened emails is 16.45%; 15.14% of clicked emails lead to introductions.

Economic message. Clicks are meaningful because they convert into introductions and investments.

6.3 Table II: Descriptive Statistics for Start-Ups

Read: The 21 firms are concentrated in technology and consumer sectors and have meaningful team, advisor, employee, and prior funding information.

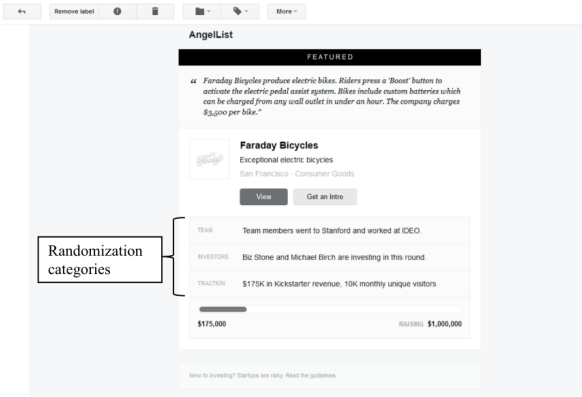


Figure 1. Sample featured start-up email to investors. This figure shows an example of a featured start-up email that is sent to investors. Each featured start-up has up to three information categories (team, traction, and current investors) that would normally be shown in the email. The information for that category reaches a threshold as defined by AngelList. For each start-up, various unique versions of each email are generated that randomly hide these pieces of information (the *Randomization categories*). Each email contains a “View” button that, when clicked, takes the investor to the AngelList platform where more information about the company is shown and introductions to the company’s founders can be requested. The “Get an Intro” button requests a straight introduction from the email.

or MIT, or if they worked at a top company such as Google or PayPal prior to starting the company. The final piece of information, which is always shown, is the amount the company aims to raise and its progress toward that goal.

Sample featured start-up email to investors

Table 1

Descriptive Statistics of Emails for Randomized Field Experiment

Panel A reports descriptive statistics for the emails about featured start-ups in the experiment. Each start-up has up to three information categories (team, traction, and current investors), each of which would usually be shown if the category passes AngelList’s disclosure threshold. For each start-up, various unique emails are generated that randomly hide these pieces of information. This sample is limited to emails to active investors who have requested at least one introduction in the past. *Clicked* measures the frequency with which investors click on the button in the email that links to the AngelList platform, where more information is shown and introductions can be requested. *Intro* means an investor requested an introduction within three days of viewing the email. *Investment* means the investor invested in the company at any time after receipt of the email. Panel B reports the frequency with which each category passed the disclosure threshold for 160 companies featured prior to the experiment (first line), and for the 21 experiment firms (second line). The third line shows the *p*-values of a test with null hypothesis of equal frequencies across the two samples, for each information category. The last line shows how often the information was actually shown in the experiment emails conditional on having passed the threshold. The rightmost column shows the *p*-value for the Pearson’s χ^2 test of the null hypothesis that the proportions in the first three columns are all equal.

Panel A: Experiment Descriptive Statistics					
	Mean	SD	Percentile		
			10	50	90
Emails					
Total	16,981				
Unique	58				
Investors/unique email	293	149	86	264	468
Active investors emailed	4,494				
Active investors who opened at least one email	2,925				
Start-ups	21				
Investors/start-up	809	468	338	676	1,457
Start-ups/investor	3.78	2.45	1	3	7
Emails opened (%)	48.22				
Clicked (% of opened emails)	16.45				
Intro (% of clicked emails)	15.14				
Investment (% of clicked emails)	2.98				
Panel B: Information in Emails					
	Team	Investors	Traction	<i>p</i> -Value	
Information passed threshold					
Preexperiment (160 firms; % of start-ups)	89.38	80.00	83.13	0.065	
Experiment (21 firms; % of start-ups)	90.48	80.95	85.71	0.678	
Equality of proportions test <i>p</i> -value	0.877	0.918	0.764		
Information shown in experiment, if passed threshold (% of unique emails)	73.24	73.02	72.06	0.987	

thresholds 89.4%, 80.0%, and 83.3% of the time, respectively. The second line is

Email descriptive statistics

Economic message. These are credible early-stage firms, not hypothetical projects.

Table II
Descriptive Statistics for Start-Ups

This table reports descriptive statistics for the 21 start-ups in the randomized field experiment at the time of fund-raising. Panel A reports the distribution across cities and countries. Panel B reports the distribution across sectors, where sectors are not mutually exclusive. Panel C reports the structure of the start-up in terms of number of founders, employees, board members, advisors, and whether the company has an attorney. *Employees (%)* is the fraction of start-ups that have nonfounder employees. The *If > 0*, the # *employees* variable shows how many employees are working for those start-ups that have employees. The variables for board members, advisors, an attorney follow a similar pattern. Panel D reports the percentage of start-ups that had funding prior to the current round (*Preround funding (%)*) and, if any prior money was raised, the amount raised (*If > 0, preround funding raised*). *Incubator (%)* is the fraction of start-ups that have been part of an incubator or accelerator program in the past, and *Equity financing (%)* is the percentage of firms selling stock, with the remainder selling convertible notes.

Panel A: Start-Up Distribution Across Cities						
	<i>N</i>	Fraction (%)				
Austin, TX	1	4.76				
Chicago, IL	1	4.76				
Kitchener, Canada	1	4.76				
London, United Kingdom	1	4.76				
Melbourne, Australia	1	4.76				
New York City, NY	3	14.28				
San Antonio, TX	1	4.76				
Silicon Valley, CA	6	28.57				
Singapore	1	4.76				
Sydney, Australia	1	4.76				
Toronto, Canada	3	14.28				
Vancouver, Canada	1	4.76				
Panel B: Start-Up Distribution Across Sectors						
	<i>N</i>	Fraction (%)				
Information technology	18	85.71				
Consumers	13	61.90				
Clean technology	1	4.76				
Healthcare	3	14.28				
Business-to-business	8	38.10				
Media	2	9.52				
Education	2	9.52				
Panel C: Start-Up Structure						
	<i>N</i>	Mean	<i>SD</i>	Percentile		
				10	50	90
# Founders	21	2.62	0.92	2	2	4
Employees (%)	21	80.95				
If > 0, # employees	17	3.35	2.21	1	3	7
Board members (%)	21	23.81				
If > 0, # board members	5	1.80	0.84	1	2	5
Advisor (%)	21	90.48				
If > 0, # advisors	19	4.74	6.00	1	3	7
Attorney (%)	21	71.43				

(a) Panels A–C

Table II—Continued						
Panel D: Start-Up Funding						
	<i>N</i>	Mean	<i>SD</i>	Percentile		
				10	50	90
Incubator (%)	21	57.14				
Preround funding (%)	21	52.38				
If > 0, amount raised (\$000s)	11	580.95	855.33	50	290	950
Premoney valuation (\$000s)	16	5,465.63	2,133.60	3,000	5,000	8,000
Fund-raising goal (\$000s)	18	1,183.06	462.88	570	1,250	2,000
Equity financing (%)	21	76.19				

(b) Panel D

Start-up descriptive statistics

6.4 Table III: Descriptive Statistics for Investors

Read: Investors are experienced, connected, and often involved as founders, advisors, board members, or investors.

Economic message. The response sample is a sophisticated investor population.

Table III
Descriptive Statistics for Investors

This table reports descriptive statistics for the active investors (defined as having requested a least one introduction through the AngelList platform) who received featured emails about th start-ups in the randomized field experiment and opened at least one such email. Panel A report sectors in which investors have stated that they are interested in investing. A single investor ca indicate multiple sectors of interest. Panel B reports the number of introductions requested b investors, the signal of an investor's success as computed by AngelList (see the main text for description of the algorithm), the number of followers that investors have on the platform (both th raw number and weighted by the followers' signal), the percentage of investors that were involve with start-ups in the past, and for those involved with start-ups, and the number of start-up the investor was involved with. Panel C breaks down involvement into various roles. *Investor (%)* is the percentage of investors who have invested in start-ups. For the subset of investors wh invested in start-ups, *If > 0, # start-ups funded* reports the number of start-ups that they investe in. The variable definitions for advisor, board member, and founder follow a similar pattern. Pane D reports the correlations between the number of investments, signal, and number of followers.

Panel A: Investor Stated Interest Across Sectors						
Sector	<i>N</i>	Fraction (%)				
Information technology	2,884	98.59				
Consumers	2,769	94.66				
Clean technology	861	29.43				
Healthcare	1,239	42.35				
Business-to-business	2,328	79.58				
Finance	949	32.44				
Media	1,420	48.54				
Energy	165	5.64				
Education	685	23.41				
Life sciences	414	14.15				
Transportation	307	10.49				
Other	26	0.80				

Panel B: Investor Characteristics						
	<i>N</i>	Mean	<i>SD</i>	Percentile		
				10	50	90
# Introductions requested	2,925	9.72	31.09	1	3	21
Signal	2,925	6.44	2.26	3.28	6.30	9.87
# Followers	2,925	591.12	1,493.10	26	202	1,346
Weighted # followers	2,925	2,527.30	5,763.70	108.97	915.70	5,896.9
Involved in start-ups (%)	2,925	91.93				
If > 0, # start-ups involved with	2,689	12.55	17.18	2	8	27

(a) Panels A–B

Table III—Continued						
Panel C: Investor Roles in Start-Up Companies						
	<i>N</i>	Mean	<i>SD</i>	Percentile		
				10	50	90
Investor (%)	2,925	82.36				
If > 0, # start-ups funded	2,409	13.10	16.81	2	8	28
Advisor (%)	2,925	43.49				
If > 0, # start-ups as advisor	1,272	3.47	4.54	1	2	7
Board member (%)	2,925	16.92				
If > 0, # start-ups as board member	495	1.93	1.82	1	1	4
Start-up founder (%)	2,925	60.00				
If > 0, # start-ups founded	1,755	2.05	1.44	1	2	4

Panel D: Correlations between Investor Heterogeneity Measures				
	Number of Investments	Signal	# Followers	Weighted # Followers
Number of investments	1			
Signal	0.51	1		
# Followers	0.63	0.41	1	
Weighted # followers	0.68	0.44	0.95	1

(b) Panels C–D

Investor descriptive statistics

6.5 Table IV: Investor Response to Randomized Emails

Read: Team disclosure raises click probability by about 2.2–2.3 percentage points; current investors and traction are not significant.

Economic message. The headline causal result is that investors respond to founding-team information.

6.6 Table V: Investor Response to Nonrandomized Emails

Read: Nonrandomized comparisons make team and traction appear important; randomized estimates isolate team.

Economic message. Randomization prevents omitted start-up quality from being mistaken for treatment effects.

6.7 Table VI: Response of Specialized Investors

Read: Specialized investors respond similarly to team information.

Table IV
Investor Response to Randomized Emails

This table reports regression results of investor responses to the featured emails in the randomize field experiment. The dependent variable is one when an investor clicked on the "View" button of the featured email, and zero otherwise. Only opened emails are included in the sample. *Team = 1* is an indicator variable that equals one if the team information is shown in the email, and zero otherwise. Similarly, *Investors = 1* and *Traction = 1* are indicator variables for current investor and traction information, respectively. *Connections* count the number of people on the start-up profile (in any role) that the investor already follows prior to receiving the email. *Prior follow = 1* is an indicator variable that equals one if the investor was already following the start-up on AngelList prior to receiving the featured email. *Prior emails* is the number of emails that the investor has received in the experiment prior to the present email. R^2 is the adjusted R^2 for OLS regressions, and pseudo R^2 for logit models. Standard errors are in parentheses and are clustered at the investor level. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	OLS (1)	OLS (2)	Logit (3)	Logit (4)
Team = 1	0.022** (0.010)	0.023** (0.010)	0.162** (0.073)	0.172** (0.074)
Investors = 1	0.010 (0.013)	0.009 (0.013)	0.070 (0.097)	0.067 (0.097)
Traction = 1	0.016 (0.014)	0.017 (0.014)	0.122 (0.106)	0.123 (0.106)
Connections		0.010 (0.006)	0.064* (0.038)	
Prior follow = 1		0.143*** (0.033)	0.835** (0.166)	
Prior emails		0.001 (0.003)	0.006 (0.022)	
Start-up fixed effects	Y	Y	Y	Y
Number of observations	8,189	8,189	8,189	8,189
R^2	0.001	0.005	0.028	0.033

IV. Main Results

Investor response to randomized emails

Economic message. Team information is not merely a signal about the business idea; it likely reflects operational human capital.

Table VI
Response of Specialized Investors

This table reports regression results of investor responses to the featured emails in the randomize field experiment. The dependent variable is one when an investor clicked on the "View" button of the featured email, and zero otherwise. Only opened emails are included in the sample. *Team = 1*, *Investors = 1*, and *Traction = 1* are indicator variables that equal one if the team, current investor or traction information, respectively, are shown in the email. *Similarity = 1* is an indicator variable that equals one if the distance between the vectors capturing investor sector interests and the start up's sectors is at the top 25% of the distance distribution. See the main text for the algorithm used to compute the distance metric. *Other controls* include *Connections*, *Prior follow*, and *Prior emails* as defined in Table IV, but are not reported for brevity. R^2 is the adjusted R^2 for OLS regression; and pseudo R^2 for logit models. Standard errors are in parentheses and are clustered at the investor level. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	OLS (1)	OLS (2)	OLS (3)	OLS (4)	Logit (5)	Logit (6)	Logit (7)	Logit (8)
Team = 1	0.023** (0.010)	0.024** (0.010)	0.024** (0.011)	0.023** (0.011)	0.172** (0.074)	0.177** (0.074)	0.190** (0.085)	0.177* (0.087)
Investors = 1	0.009 (0.013)	0.009 (0.013)	0.009 (0.013)	0.003 (0.014)	0.067 (0.097)	0.068 (0.097)	0.068 (0.097)	0.014 (0.108)
Traction = 1	0.017 (0.014)	0.016 (0.014)	0.016 (0.014)	0.011 (0.016)	0.123 (0.106)	0.119 (0.106)	0.120 (0.117)	0.090 (0.117)
Similarity = 1		0.039*** (0.011)	0.041*** (0.015)	0.006 (0.035)		0.291*** (0.079)	0.315*** (0.112)	0.081 (0.260)
Similarity = 1 × Team = 1			-0.002 (0.019)	0.004 (0.020)			-0.041 (0.136)	0.002 (0.143)
Similarity = 1 × Investors = 1				0.026 (0.023)				0.192 (0.167)
Similarity = 1 × Traction = 1				0.017 (0.025)				0.099 (0.182)
Start-up fixed effects	Y	Y	Y	Y	Y	Y	Y	Y
Other controls	Y	Y	Y	Y	Y	Y	Y	Y
Number of observations	8,189	8,189	8,189	8,189	8,189	8,189	8,189	8,189
R^2	0.005	0.007	0.007	0.007	0.033	0.035	0.035	0.035

Response of specialized investors

Table V
Investor Response to Nonrandomized Emails

This table replicates the regressions in Table IV for the subset of featured emails that show a information that has crossed the disclosure threshold, in the columns labeled "Full Information Subsample." The model numbers in the second row correspond to the model numbers in Table IV. For ease of comparison, the columns labeled "Randomized Sample" show the results from Table I for the same set of models. The dependent variable is one when an investor clicked on the "View" button in the featured email, and zero otherwise. The explanatory variables are as defined in Table IV. R^2 is the adjusted R^2 for OLS regressions, and pseudo R^2 for logit models. Standard errors are in parentheses and are clustered at the investor level. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	Full Information Subsample				Randomized Sample			
	OLS (1)	OLS (2)	Logit (3)	Logit (4)	OLS (1)	OLS (2)	Logit (3)	Logit (4)
Team = 1	0.046** (0.022)	0.045** (0.022)	0.336** (0.171)	0.337* (0.172)	0.022** (0.010)	0.023** (0.010)	0.162** (0.073)	0.172** (0.074)
Investors = 1	0.013 (0.018)	0.022 (0.019)	0.091 (0.127)	0.155 (0.133)	0.010 (0.013)	0.009 (0.013)	0.070 (0.097)	0.067 (0.097)
Traction = 1	0.037* (0.020)	0.043** (0.020)	0.265* (0.149)	0.311** (0.154)	0.016 (0.014)	0.017 (0.014)	0.122 (0.106)	0.123 (0.106)
Connections		0.010 (0.010)		0.058 (0.054)		0.010 (0.006)		0.064* (0.038)
Prior follow = 1		0.150** (0.059)		0.822*** (0.277)		0.143*** (0.033)		0.835** (0.166)
Prior emails		-0.006* (0.003)		-0.042* (0.025)		0.001 (0.003)		0.006 (0.022)
Start-up fixed effects	N	N	N	N	Y	Y	Y	Y
Number of observations	2,992	2,992	2,992	2,992	8,189	8,189	8,189	8,189
R^2	0.001	0.006	0.002	0.008	0.001	0.005	0.028	0.033

that no investor received more than one email for any one featured start-up

Investor response to nonrandomized emails

Table VII
Investor Response by Number of Investments

This table reports regression results of investor responses to the featured emails in the randomize field experiment. The dependent variable is one when an investor clicked on the "View" button of the featured email, and zero otherwise. Only opened emails are included in the sample. *Team = 1*, *Investors = 1*, and *Traction = 1* are indicator variables that equal one if the team, current investor or traction information, respectively, are shown in the email. *# Investments ≤ cutoff* is an indicator variable that equals one if the number of investments by a given investor is less than or equal to the percentile of the investments count distribution shown in the row labeled *Cutoff*. The other variables are as defined in Table IV. R^2 is the adjusted R^2 for OLS regressions, and pseudo R^2 for logit models. Standard errors are in parentheses and are clustered at the investor level. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	OLS (1)	OLS (2)	OLS (3)	Logit (4)	Logit (5)	Logit (6)
Team = 1	0.017* (0.010)	0.021* (0.011)	0.026** (0.013)	0.130* (0.079)	0.162* (0.087)	0.203** (0.099)
Investors = 1	-0.001 (0.013)	0.004 (0.014)	0.003 (0.016)	-0.012 (0.104)	0.025 (0.115)	0.016 (0.125)
Traction = 1	0.009 (0.014)	0.003 (0.016)	0.010 (0.017)	0.066 (0.108)	0.024 (0.117)	0.076 (0.129)
# Investments ≤ cutoff × Team = 1	0.037 (0.025)	0.007 (0.018)	-0.007 (0.017)	0.235 (0.176)	0.029 (0.137)	-0.061 (0.131)
# Investments ≤ cutoff × Investors = 1	0.070** (0.028)	0.021 (0.021)	0.014 (0.020)	0.476** (0.189)	0.146 (0.151)	0.103 (0.149)
# Investments ≤ cutoff × Traction = 1	0.063** (0.031)	0.047** (0.021)	0.015 (0.020)	0.427* (0.220)	0.351** (0.163)	0.116 (0.154)
# Investments ≤ cutoff	-0.080* (0.043)	-0.028 (0.031)	-0.001 (0.030)	-0.530* (0.320)	-0.193 (0.241)	0.001 (0.231)
Connections	0.010 (0.006)	0.010 (0.006)	0.010 (0.006)	0.066* (0.038)	0.068* (0.038)	0.067* (0.038)
Prior follow	0.145*** (0.033)	0.144*** (0.033)	0.145*** (0.033)	0.847*** (0.166)	0.849*** (0.166)	0.852** (0.166)
Prior emails	0.002 (0.003)	0.002 (0.003)	0.001 (0.003)	0.013 (0.022)	0.014 (0.022)	0.010 (0.022)
Start-up fixed effects	Y	Y	Y	Y	Y	Y
Cutoff	Zero	25%	50%	Zero	25%	50%
Number of observations	8,189	8,189	8,189	8,189	8,189	8,189
R^2	0.007	0.006	0.005	0.035	0.035	0.034

Response by number of investments

6.8 Tables VII–IX: Investor response by experience, signal, and weighted followers

Read: Experienced and successful investors focus primarily on team information.

Economic message. This is indirect evidence that team-based screening is a viable investment strategy.

Table VIII
Investor Response by Signal

This table reports regression results of investor responses to the featured emails in the randomized field experiment. The dependent variable is one when an investor clicked on the “View” button in the featured email, and zero otherwise. Only opened emails are included in the sample. $Team = 1$, $Investors = 1$, and $Traction = 1$ are indicator variables that equal one if the team, current investors, or traction information, respectively, are shown in the email. $Signal < cutoff$ is an indicator variable that equals one if the investor signal is below the percentile of the signal distribution shown in the row labeled $Signal < cutoff$. See the main text for the algorithm used to compute the signals. The other variables are as defined in Table IV. R^2 is the adjusted R^2 for OLS regressions, and pseudo R^2 for logit models. Standard errors are in parentheses and are clustered at the investor level. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

	OLS (1)	OLS (2)	OLS (3)	Logit (4)	Logit (5)	Logit (6)
Team = 1	0.019* (0.011)	0.024* (0.013)	0.035* (0.019)	0.147* (0.085)	0.184* (0.097)	0.265* (0.139)
Investors = 1	-0.003 (0.014)	0.001 (0.016)	-0.005 (0.022)	-0.036 (0.112)	0.005 (0.127)	-0.045 (0.170)
Traction = 1	0.005 (0.015)	0.010 (0.017)	0.009 (0.022)	0.037 (0.112)	0.077 (0.123)	0.067 (0.156)
Signal < cutoff	0.013 (0.020)	-0.003 (0.017)	-0.016 (0.020)	0.067 (0.145)	-0.027 (0.131)	-0.127 (0.154)
× Team = 1	0.055** (0.022)	0.016 (0.020)	0.019 (0.024)	0.385** (0.159)	0.121 (0.151)	0.141 (0.184)
Signal < cutoff	0.063** (0.024)	0.015 (0.020)	0.012 (0.023)	0.440** (0.180)	0.122 (0.155)	0.093 (0.171)
× Investors = 1	-0.049 (0.034)	-0.013 (0.030)	0.001 (0.036)	-0.320 (0.261)	-0.102 (0.231)	0.010 (0.276)
Signal < cutoff	0.011* (0.006)	0.010 (0.006)	0.010 (0.006)	0.075* (0.038)	0.066* (0.038)	0.067* (0.038)
Connections	0.144*** (0.033)	0.144*** (0.033)	0.145*** (0.033)	0.843*** (0.166)	0.841*** (0.166)	0.846*** (0.166)
Prior follow	0.003 (0.003)	0.001 (0.003)	0.001 (0.003)	0.020 (0.022)	0.008 (0.022)	0.006 (0.022)
Prior emails	Y	Y	Y	Y	Y	Y
Start-up fixed effects	25%	50%	75%	25%	50%	75%
Cutoff	8,189	8,189	8,189	8,189	8,189	8,189
Number of observations	8,189	8,189	8,189	8,189	8,189	8,189
R^2	0.007	0.005	0.005	0.036	0.033	0.034

Response by signal

Table IX
Investor Response by Weighted Number of Followers

This table reports regression results of investor responses to the featured emails in the randomized field experiment. The dependent variable is one when an investor clicked on the “View” button in the featured email, and zero otherwise. Only opened emails are included in the sample. $Team = 1$, $Investors = 1$, and $Traction = 1$ are indicator variables that equal one if the team, current investors, or traction information, respectively, are shown in the email. $Wtd \# followers < cutoff$ is an indicator variable that equals one if the number of followers of a given investor, weighted by their signal, is less than the percentile of the weighted followers count distribution shown in the row labeled $Cutoff$. The other variables are as defined in Table IV. R^2 is the adjusted R^2 for OLS regressions, and pseudo R^2 for logit models. Standard errors are in parentheses and are clustered at the investor level. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively.

	OLS (1)	OLS (2)	OLS (3)	Logit (4)	Logit (5)	Logit (6)
Team = 1	0.020* (0.011)	0.026** (0.013)	0.033* (0.017)	0.158* (0.085)	0.207** (0.104)	0.268* (0.140)
Investors = 1	-0.003 (0.013)	-0.017 (0.016)	-0.013 (0.020)	-0.036 (0.109)	-0.156 (0.127)	-0.111 (0.164)
Traction = 1	0.006 (0.015)	0.003 (0.017)	0.014 (0.021)	0.048 (0.113)	0.023 (0.126)	0.112 (0.158)
Wtd # followers < cutoff	0.009 (0.020)	-0.008 (0.017)	-0.014 (0.020)	0.035 (0.144)	-0.082 (0.132)	-0.129 (0.157)
× Team = 1	0.051** (0.024)	0.051*** (0.020)	0.028 (0.022)	0.354** (0.165)	0.399*** (0.152)	0.221 (0.176)
Wtd # followers < cutoff	0.049** (0.024)	0.030 (0.020)	0.003 (0.022)	0.338* (0.177)	0.229 (0.157)	0.023 (0.176)
× Investors = 1	-0.035 (0.034)	-0.018 (0.029)	0.021 (0.034)	-0.216 (0.254)	-0.117 (0.231)	0.183 (0.276)
Wtd # followers < cutoff	0.013** (0.006)	0.013** (0.006)	0.013** (0.006)	0.083** (0.039)	0.090** (0.040)	0.084** (0.039)
Connections	0.145*** (0.033)	0.146*** (0.033)	0.145*** (0.033)	0.855*** (0.166)	0.861*** (0.166)	0.854*** (0.166)
Prior follow	0.002 (0.003)	0.001 (0.003)	0.001 (0.003)	0.018 (0.022)	0.008 (0.022)	0.005 (0.021)
Prior emails	Y	Y	Y	Y	Y	Y
Start-up fixed effects	25%	50%	75%	25%	50%	75%
Cutoff	8,189	8,189	8,189	8,189	8,189	8,189
Number of observations	8,189	8,189	8,189	8,189	8,189	8,189
R^2	0.007	0.008	0.008	0.036	0.033	0.035

Response by weighted followers

6.9 Table X: Start-Ups in Field Experiment Sample versus Broad Sample

Read: Experiment firms are somewhat stronger on founders, employees, advisors, and incubator history, but not on every funding variable.

Economic message. External validity is strongest for selected early-stage firms attractive enough to be featured.

Table X
Start-Ups in Field Experiment Sample versus Broad Sample

This table compares the sample of 21 start-ups in the randomized field experiment (“experiment firms”) with a broad sample of 5,538 firms raising funds on AngelList (“nonexperiment firms”). The nonexperiment firms are those firms that attempted to raise money through AngelList and received at least one introduction request. The variables are as defined in Table II. The rightmost column shows the p -value for a difference-in-means test between the experiment and nonexperiment samples. For nonpercentage numbers, the table shows means, with medians in brackets.

	Experiment Firms ($N = 21$)			Nonexperiment Firms ($N = 5,538$)			Means Test
	N	Mean	SD	N	Mean	SD	p -Value
# Founders	21	2.62 (2)	0.92	5,538	2.11 (2)	1.06	0.028
Employees (%)	21	80.95		5,538	52.56		0.009
If > 0, # employees	17	3.35 (3)	2.21	2,911	2.91 (2)	2.45	0.453
Board members (%)	21	23.81		5,538	16.78		0.390
If > 0, # board members	5	1.80 (2)	0.84	929	1.96 (2)	1.14	0.749
Advisor (%)	21	90.48		5,538	60.74		0.005
If > 0, # advisors	19	4.74 (3)	6.00	3,364	2.94 (2)	2.18	0.000
Incubator (%)	21	57.14		5,538	29.70		0.006
Preround funding (%)	21	47.62		5,538	45.76		0.865
If > 0, amount raised (\$000s)	10	605.05 (234.00)	897.66	2,534	674.27 (250.00)	1,874.28	0.904
Premoney valuation (\$000s)	12	5,579.17 (5,000.00)	2,383.22	2,616	4,857.83 (3,500.00)	15,747.91	0.873
Fund-raising goal (\$000s)	15	1,226.33 (1,325.00)	488.96	4,321	923.99 (500.00)	1,135.56	0.303
Equity financing (%)	21	76.19		4,912	69.04		0.603

Experiment sample versus broad AngelList sample

7 Interpretive synthesis

- Human capital is central when firms are too young to have extensive hard assets or performance histories.
- Traction and current investors are not irrelevant, but in this email environment they do not drive causal investor attention.
- The focus of experienced investors on team information suggests team-based screening is not simply a behavioral attention bias.

8 Short summary

- The paper randomizes information in real AngelList emails.
- Investors respond strongly to team disclosure.
- Randomized estimates do not show significant responses to traction or current-investor disclosure.
- Specialized-investor results suggest the team matters beyond signaling the idea.
- Experienced and successful investors focus on the team.

9 Conclusion

9.1 Core conclusion

The paper shows that early-stage investors are causally attracted to founding-team information. In a randomized field experiment on AngelList, showing team information raises investor click-through rates, while showing traction or current-investor information does not generate comparable average responses.

Interpretation. The conclusion should be read as a statement about the earliest screening stage of financing. The paper demonstrates what moves investor attention when investors first evaluate a start-up in a low-information, high-uncertainty setting.

9.2 What the paper establishes

- **Human capital is salient and causally important.** Investors are more likely to investigate a start-up when the founding team is disclosed.
- **The effect is not just observational sorting.** Because disclosure is randomized within start-up, the result is not explained by strong teams being attached to better ideas.
- **Team information is not only a naive-investor signal.** Specialized, experienced, and successful investors place particular weight on team information.
- **Clicks are economically meaningful.** They are the first measurable step toward introduction requests and potential investment.
- **Nonhuman assets still matter later.** The paper does not claim that traction, technology, customers, or investor networks are irrelevant; it shows that they are not the strongest causal disclosure margins in this early screening environment.

9.3 What the paper does not establish

- It does not estimate the causal effect of team quality on final start-up success.
- It does not prove that team-focused investors earn higher realized returns.
- It does not show that traction and current investors are unimportant in all fundraising settings.
- It does not fully generalize to later-stage VC rounds, offline negotiation, or firms outside platform-based seed financing.
- It does not separate every possible dimension of team quality, such as education, entrepreneurial experience, technical background, or social capital.

Interpretation. These limits are not weaknesses in the experimental design; they define the paper’s estimand. The experiment cleanly identifies the effect of information visibility on investor screening attention.

9.4 Economic intuition

Start-ups are difficult to evaluate because they have few hard assets, short operating histories, and high uncertainty. In that environment, the founding team may be the most portable and interpretable signal of execution ability. Investors may infer that a strong team can adapt, recruit, raise follow-on capital, and pivot when initial plans fail. Traction and current-investor signals can matter, but they may be noisier or more context-specific at the seed stage.

9.5 Connection to theories of the firm

The paper connects directly to competing theories of what makes a young firm valuable. Human-capital theories predict that the people behind the firm are central at the earliest stage. Property-rights or nonhuman-asset views place more weight on the underlying idea, technology, customers, or traction. Information-cascade views predict strong responses to existing backers. The evidence favors the human-capital channel in the initial screening environment.

Interpretation. The result is consistent with a life-cycle view of entrepreneurial finance: human capital may differentiate the firm early, while standardized assets, product-market traction, and replaceable organizational routines become more important later.

9.6 Practical implications

- **For founders.** Early-stage pitches should communicate credible, concrete team information, especially evidence of skills, execution history, and relevant experience.
- **For platforms.** Search and recommendation systems should treat team disclosure as a high-value screening input, but should avoid overstating what clicks imply about ultimate success.
- **For investors.** Team-based screening appears to be used by sophisticated investors, but it should be combined with deeper diligence on product, market, financing terms, and follow-on potential.
- **For researchers.** The paper shows how field experiments can identify investor information preferences in private capital markets where standard observational data are incomplete.

9.7 Incremental takeaway

The incremental contribution of the paper is to move the entrepreneurial-finance literature from stated preferences and observational correlations to causal evidence on investor screening. It shows that, when the same start-up is held fixed and only information visibility changes, investors respond most strongly to the founding team. This makes human capital a measurable causal input into early-stage fundraising attention.

9.8 Final takeaway

In early-stage finance, investors appear to fund people before products. The paper’s central message is not that products, traction, or backers do not matter, but that the first causal margin of investor attention is the founding team. Human capital is therefore a central asset in the earliest stage of firm development.